

PROBLEM STATEMENTS

Q1. Hybrid AI Recommendation Engine Design

Traditional recommendation systems often rely on only one technique, such as content-based filtering or collaborative filtering, which may not provide accurate recommendations for all users. The proposed system combines content-based and collaborative filtering using PySpark RDDs to process large-scale user-item interactions in a distributed manner. Intelligent agents monitor user behavior and feedback in real time to adapt and improve personalized recommendations.

Q2. Smart Disaster Detection System

Natural disasters such as **floods and earthquakes** require early and accurate detection. However, information is generated from multiple sources such as **satellite images, sensors, and social media**, making it difficult to process and combine the data efficiently. The proposed system uses **PySpark and RDD operations** to process multi-source data in a distributed manner, apply data fusion techniques, and use intelligent agents to coordinate disaster detection, early warnings, and response strategies.

Q3. AI-Based Fraud Detection Framework

The increasing volume of financial transactions makes it difficult to manually identify fraudulent activities. Fraudsters can also create new and unusual transaction patterns that are difficult to detect using traditional methods. The proposed system uses **PySpark RDDs** to process large volumes of financial transactions and apply distributed **anomaly detection**. Intelligent agents collaborate to identify suspicious patterns, generate alerts, support investigations, and continuously improve fraud detection accuracy.

Q4. Autonomous Supply Chain Optimization System

Modern supply chains generate large amounts of data related to **inventory, logistics, sales, suppliers, and demand**. Managing this information efficiently is challenging, especially when demand and market conditions change frequently. The proposed system uses **PySpark and RDD transformations** for large-scale data processing and intelligent agents for demand forecasting, inventory management, logistics optimization, and procurement. These agents work

together to make decentralized decisions that reduce costs and improve supply-chain efficiency.

Q5. Personalized Learning Analytics Platform

Traditional educational systems may provide the same learning content to many students without considering their individual performance and learning behavior. With thousands of learners, it is challenging to analyze student data and provide personalized support in real time. The proposed platform uses **PySpark and RDDs** to analyze student learning behavior and performance data. Intelligent agents provide adaptive learning, personalized content recommendations, and real-time feedback to improve the learning experience for individual students.